BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 134 · Network Hardware and Topologies

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What distinguishes a WAN?

Q2. Why use shared computing standards?
A. Every device needs one maker
B. Different products work together
C. No communication rules are needed
D. All hardware becomes identical

Q3. “SMTP's role is synchronising folders and read status across mail clients.” Correct this. Explain why.

Q4. A star LAN has one central switch. “One broken workstation cable stops the whole network.” Correct this. Compare a broken cable with a failed switch.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

NIC — Provides a device's connection to a network.

Switch — Forwards frames between devices within a LAN using MAC address information.

Router — Connects networks and forwards packets using IP address/routing information.

Star and mesh — Star connects devices via a central point; mesh offers multiple potential paths between nodes.

New learning

NIC: Provides a device's connection to a network.

Switch: Forwards frames between devices within a LAN using MAC address information.

Router: Connects networks and forwards packets using IP address/routing information.

Star and mesh: Star connects devices via a central point; mesh offers multiple potential paths between nodes.

Worked example

Explain how data travels from a classroom PC to a server on another network.

1. The PC connects via its NIC.

2. A switch forwards traffic inside the LAN.

3. A router forwards the packet to the other network.

4. The destination network delivers data to the server.

We Do · Practise with guidance

Guided 1

Compare star and mesh topology after one connection fails.

Worked method: Explain the trade-off: mesh uses more connections and costs more.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the number of links in a fully connected mesh of five computers.

```
devices = 5
links = devices * (devices - 1) DIV 2
print(links)
```

predict

Question 1

State what a switch does in a LAN. [2 marks]

Question 2

Give the purpose of a router. [2 marks]

Question 3

Calculate full-mesh connections for four nodes. [2 marks]

Exit Ticket · Answer before checking

A switch normally forwards using IP routes between networks.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A WAN links networks over a large area, often using links provided by other organisations.

Q2. B — Different products work together

Q3. Sending email from a client to a mail server and between mail servers.

Q4. A broken workstation cable usually affects just that device. If the central switch fails, devices using it lose contact. Available paths determine which connections still work.

I Do / We Do pages · Guided practice

Q1. Explain the trade-off: mesh uses more connections and costs more.

You Do pages · Code First

```
devices = 5
links = devices * (devices - 1) DIV 2
print(links)
```

Expected result: 10

You Do pages · Independent questions

Q1. Forwards frames to the appropriate device/port using learned MAC address information.

Q2. Connects separate networks and forwards packets between them using IP routing.

Q3. 4×3÷2=6 connections.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: